

📌 Part 1: Introduction to Naive Bayes

◆ What is Naive Bayes?

👉 Naive Bayes is a Supervised Machine Learning algorithm mainly used for **Classification** problems.

👉 It predicts a category based on probability.

Examples:

- Spam Detection
 - Email Classification
 - Sentiment Analysis
 - Disease Prediction
-

◆ Why is it Called "Naive Bayes"?

The algorithm is based on:

1. Bayes Theorem
 2. A simple (naive) assumption that features are independent
-

📌 Part 2: Real-Life Example

Imagine you receive an email containing words like:

Free
Offer
Win
Prize

Most spam emails contain these words.

So the algorithm calculates:

👉 What is the probability that this email is spam?

If the probability is high:

✓ Spam

Otherwise:

✓ Not Spam

✂ Part 3: Theory Answers (IMPORTANT)

◆ 1. What is Naive Bayes?

👉 Naive Bayes is a probability-based supervised learning algorithm used for classification.

◆ 2. Which theorem is used in Naive Bayes?

👉 Bayes Theorem.

◆ 3. Is Naive Bayes used for Classification or Regression?

👉 Classification.

◆ 4. What is Probability?

👉 Probability is the chance of an event occurring.

Example:

Chance of rain tomorrow.

◆ 5. What are the common applications of Naive Bayes?

👉 Spam filtering, sentiment analysis, document classification, and disease prediction.

✂ Part 4: Understanding Probability

Suppose:

100 emails

Type	Count
------	-------

Spam	70
------	----

Not Spam	30
----------	----

Probability of spam:

$$70 / 100 = 0.70$$

Meaning:

👉 70% chance email is spam.

📌 Part 5: Bayes Theorem (Basic Understanding)

Naive Bayes uses Bayes Theorem.

The idea is:

New Data



Calculate Probability



Choose Highest Probability



Prediction

📌 Part 6: Example Dataset

Study Hours Result

1	Fail
---	------

2	Fail
---	------

3	Fail
---	------

5	Pass
---	------

6	Pass
---	------

7	Pass
---	------

Part 7: Import Libraries

```
import pandas as pd
```

```
from sklearn.naive_bayes import GaussianNB
```

Part 8: Create Dataset

```
data = {  
    "Hours": [1, 2, 3, 5, 6, 7],  
    "Result": [0, 0, 0, 1, 1, 1]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Meaning

Value Meaning

0 Fail

1 Pass

Part 9: Features and Labels

```
X = df[["Hours"]]
```

```
y = df["Result"]
```

Part 10: Create Naive Bayes Model

```
model = GaussianNB()
```

Why GaussianNB?

 It is the most commonly used Naive Bayes classifier for numerical data.

Part 11: Train the Model

```
model.fit(X, y)
```

Part 12: Make Prediction

```
prediction = model.predict([[4]])
```

```
print(prediction)
```

Possible Output:

```
[1]
```

Meaning:

 PASS

Part 13: Predict Multiple Values

```
prediction = model.predict([[2],[4],[8]])
```

```
print(prediction)
```

Possible Output:

```
[0 1 1]
```

Part 14: Complete Program

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
```

```
data = {
    "Hours": [1,2,3,5,6,7],
    "Result": [0,0,0,1,1,1]
}
```

```
df = pd.DataFrame(data)
```

```
X = df[["Hours"]]
y = df["Result"]

model = GaussianNB()

model.fit(X, y)

prediction = model.predict([[5]])

print("Prediction:", prediction)
```

Part 15: Real-World Applications

Email Filtering

 Spam / Not Spam

Sentiment Analysis

 Positive / Negative Review

Healthcare

 Disease Detection

News Classification

 Sports / Politics / Technology

E-Commerce

 Product Recommendation Categories

Part 16: Advantages of Naive Bayes

 Simple Algorithm

 Fast Training

- ✓ Fast Prediction
 - ✓ Works well with text data
 - ✓ Good for large datasets
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✚ Part 17: Disadvantages of Naive Bayes

- ✗ Assumes features are independent
 - ✗ May not work well for complex relationships
 - ✗ Sometimes less accurate than advanced models
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✚ Part 18: Naive Bayes Workflow

Dataset



Features (X)



Probability Calculation



Highest Probability



Prediction

✚ Part 19: Naive Bayes vs KNN

Naive Bayes

KNN

Probability Based

Distance Based

Faster

Slower

Works well for text

Works well for small datasets

Model Based

Instance Based

✚ Part 20: Naive Bayes vs Decision Tree

Naive Bayes Decision Tree

Uses Probability Uses Decision Rules

Faster Easier to Visualize

Good for Text Good for Structured Data

Part 21: Why Naive Bayes is Popular?

 Many email systems use Naive Bayes because:

- Fast
- Lightweight
- Excellent for text classification

Spam filtering is one of its most famous applications.

Tasks for Day 42

Task 1: Theory

1. What is Naive Bayes?
 2. What is Probability?
 3. Which theorem is used in Naive Bayes?
 4. Advantages of Naive Bayes?
 5. Real-world applications?
-

Task 2: Dataset Creation

1. Student Result Dataset
 2. Disease Dataset
 3. Spam Dataset
-

Task 3: Model Building

1. Create Naive Bayes model

2. Train model
 3. Make predictions
-

Task 4: Prediction Practice

Predict:

1. Student Result
 2. Disease Status
 3. Spam Detection
-

Task 5: Probability Activity

Suppose:

- Total Emails = 200
- Spam Emails = 150

Calculate:

1. Probability of Spam
 2. Probability of Not Spam
-

Task 6: Compare Algorithms

Compare:

1. KNN
2. Decision Tree
3. Random Forest
4. Naive Bayes